

TerraSeg

Offroad Semantic Scene Segmentation

Duality AI · Hack the Night Hackathon

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Model
DINOv2 + ConvNeXt Head

Classes
10 Desert Categories

Metric
Mean IoU (mIoU)

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1. Introduction

1.1. Problem Statement

Autonomous off-road vehicles — known as Unmanned Ground Vehicles (UGVs) — need to understand their surroundings at every moment. To navigate safely through complex desert terrain, they rely on **semantic scene segmentation**: labeling every single pixel in a camera image with the category it belongs to (sky, rock, tree, etc.).

Without this, a UGV cannot distinguish a navigable sandy path from a rocky obstacle, or tell a dry bush from a log it might get stuck on.

1.2. The Challenge

The traditional approach requires thousands of real-world labeled images — expensive, slow, and dangerous to collect in remote deserts. This hackathon by **Duality AI** poses a different challenge:

Train a segmentation model using **synthetic** desert images from Duality AI's **Falcon** platform. Then test it on a **completely different desert location** it has never seen — measuring how well it generalizes.

1.3. Our Approach

We built **TerraSeg**, combining:

- **DINOv2** — a pre-trained Vision Transformer backbone from Meta AI, used as a frozen feature extractor
- **A lightweight ConvNeXt-style segmentation head** — a small network we train to map features to per-pixel class predictions

2. Dataset

2.1. Source

All images were generated using **FalconEditor**, Duality AI’s geospatial digital twin environment. The dataset contains synthetic RGB desert photographs with pixel-level segmentation masks.

2.2. Structure

Folder	Contents
Train/Color_Images/	RGB training photographs
Train/Segmentation/	Pixel-wise class masks
Val/Color_Images/	RGB validation photographs
Val/Segmentation/	Pixel-wise validation masks
testImages/	Unseen RGB images from a different desert — no masks

2.3. Segmentation Classes

Raw ID	Mapped ID	Class
0	0	Background
100	1	Trees
200	2	Lush Bushes
300	3	Dry Grass
500	4	Dry Bushes
550	5	Ground Clutter
700	6	Logs
800	7	Rocks
7100	8	Landscape (general ground)
10000	9	Sky

2.4. Preprocessing

Raw masks use non-sequential IDs. We remap them to 0–9 before training:

```
value_map = {0:0, 100:1, 200:2, 300:3, 500:4,  
             550:5, 700:6, 800:7, 7100:8, 10000:9}
```

All images are resized to **476** × **266** (nearest size divisible by 14 for DINOv2 patches) and normalized with ImageNet statistics.

3. Methodology

3.1. Architecture Overview

Desert Image → DINOv2 Backbone (frozen) → Patch Tokens → ConvNeXt Head (trained) → Class Mask

3.2. DINOv2 Backbone (Frozen)

DINOv2 is a Vision Transformer pre-trained by Meta AI on 140M+ images. It splits the image into **14×14 pixel patches**, encodes each into a **384-dimensional feature vector**, and uses self-attention so every patch can reference every other patch.

We keep it **completely frozen** — no weights are updated. This means faster training, better generalization, and we leverage Meta’s pre-training for free.

3.3. ConvNeXt Segmentation Head (Trained)

Layer	What it does
Reshape	Convert flat patch tokens → 2D spatial grid
Stem Conv 7×7	384 → 128 channels, large receptive field
Depthwise Conv 7×7	Spatial feature mixing per channel
Pointwise Conv 1×1	Cross-channel mixing
Classifier Conv 1×1	128 → 10 class scores per location
Bilinear Upsample	Scale back to full image resolution

500K trainable parameters total.

3.4. Training Config

Setting	Value
Epochs	10
Batch Size	2
Learning Rate	1e-4
Optimizer	SGD momentum=0.9

Loss	Cross-Entropy
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Platform	Google Colab T4 GPU
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4. Evaluation Metrics

4.1. Mean IoU (Primary — 80 pts)

$$\text{IoU}_c = \frac{\text{Predicted} \cap \text{Ground Truth}}{\text{Predicted} \cup \text{Ground Truth}}, \quad \text{mIoU} = \frac{1}{C} \sum_c \text{IoU}_c$$

IoU = 1.0 is perfect. IoU = 0.0 is completely wrong.

4.2. Dice Score

$$\text{Dice}_c = \frac{2 \times |\text{Pred} \cap \text{GT}|}{|\text{Pred}| + |\text{GT}|}$$

4.3. Pixel Accuracy

$$\text{Accuracy} = \frac{\text{Correctly classified pixels}}{\text{Total pixels}}$$

5. Results

5.1. Final Metrics

Metric	Train	Val
Loss	<i>fill</i>	<i>fill</i>
mIoU	<i>fill</i>	<i>fill</i>
Dice Score	<i>fill</i>	<i>fill</i>
Pixel Accuracy	<i>fill</i>	<i>fill</i>

5.2. Per-Class IoU

Class	IoU
Background	<i>fill</i>
Trees	<i>fill</i>
Lush Bushes	<i>fill</i>
Dry Grass	<i>fill</i>
Dry Bushes	<i>fill</i>
Ground Clutter	<i>fill</i>
Logs	<i>fill</i>
Rocks	<i>fill</i>
Landscape	<i>fill</i>
Sky	<i>fill</i>

Insert training_curves.png and per_class_metrics.png here

6. Challenges & Solutions

6.1. Non-Standard Class IDs

Problem: Raw mask values (0, 100, 200 ... 10000) are non-sequential — PyTorch needs 0 to N-1.

Fix: `convert_mask()` remaps all values before training.

6.2. Windows-Only Setup Script

Problem: `setup_env.bat` doesn't work on Linux.

Fix: Used Google Colab — platform independent, free T4 GPU, training in 30 min instead of hours.

6.3. Image Size Must Be Divisible by 14

Problem: DINOv2 patch size is 14×14 . Wrong image size breaks tokenization.

Fix: $W = \text{int}(((960/2) // 14) * 14) \rightarrow 476 \times 266$.

6.4. Class Imbalance

Problem: Sky and Landscape dominate pixels; Logs and Ground Clutter are rare.

Fix: Monitor per-class IoU separately to identify weak classes. Future work: weighted loss.

6.5. Generalization to Unseen Environments

Problem: Test images are from a different desert location.

Fix: Frozen DINOv2 provides universal features that don't overfit to one location.

7. Failure Case Analysis

7.1. Visually Similar Classes

Dry Grass vs Dry Bushes vs Landscape share similar tan/brown color palettes. The model is likely to confuse these in transitional zones.

7.2. Rare Classes

Logs and **Ground Clutter** appear in very few pixels. The model may default to predicting the surrounding dominant class instead.

7.3. Boundary Blur

Bilinear upsampling from low-res predictions back to full image size causes imprecise class boundaries — especially visible where trees meet sky.

7.4. Domain Shift

Rock formations, soil color, and vegetation density differ between training and test deserts. Novel visual patterns not seen in training may be misclassified as the nearest-looking class.

Insert comparison images from predictions/comparisons/ here to illustrate failures

8. Conclusion & Future Work

8.1. Summary

TerraSeg demonstrates that **synthetic data + pre-trained vision backbones** is a powerful combination for off-road segmentation. We trained efficiently on a free cloud GPU in under 30 minutes, while achieving generalization to unseen desert environments — exactly the real-world deployment scenario this challenge simulates.

8.2. Key Takeaways

- Synthetic data from Falcon is high quality enough to train generalizable models
- Frozen DINOv2 features transfer well to desert scene understanding
- A 500K parameter head is sufficient when the backbone is strong

8.3. Future Work

- **Weighted loss** to improve rare class (Logs, Ground Clutter) performance
- **FPN or UPerNet head** for multi-scale feature fusion and sharper boundaries
- **Data augmentation** — color jitter, flips, random crops for lighting robustness
- **Partial backbone fine-tuning** for domain-specific improvement
- **Larger DINOv2** (ViT-Base or ViT-Large) for higher feature quality

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